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In professional football the decline in high-intensity running activities from first to second half is more pronounced in players with a fast muscle typology

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Abstract

Muscle typology is heterogeneous among national level football (soccer) players, but positional differences remain unclear. Furthermore, fast typology (FT) individuals fatigue more than slow typology (ST) individuals in lab conditions. Therefore, we investigated if muscle typology is different between playing positions and if the decay in high-intensity activities from the first to the second half is larger in FT football players than in ST players. We estimated muscle typology in 147 male professional football players by measuring soleus and gastrocnemius muscle carnosine via proton magnetic resonance spectroscopy. Players were classified as ST, intermediate typology (IT) or FT and categorized as goalkeeper, center back, full back, midfielder, winger or forward. Across four seasons in-game distances covered in multiple running speed, acceleration and deceleration zones were collected during the first and second half. We found no differences in muscle typology between positions ($p = 0.412$). FT players covered 10.9% more high acceleration distance (>3 m.s^{-2}) in the first half than ST players ($p = 0.021$) and high acceleration distance decay was larger for FT players (−12.4%) than ST (−7.7%; $p = 0.006$) and IT players (−7.3%; $p = 0.010$). Moreover, the decline in distance covered in several high-intensity zones tended to be larger in FT players (−11.2% high-intensity >15 km.h^{-1}; −12.7% high deceleration <−3 m.s^{-2}; −11.5% medium acceleration 2-3 m.s^{-2}) than in ST players (−7.1% high-intensity; −8.1% high deceleration; −8.1% medium acceleration; $0.05 < p < 0.1$). In conclusion, possessing a particular muscle typology is not required to play any football position at the national level. However, there are indications that FT players might fatigue more toward the end of the game compared to ST players.

KEYWORDS

electronic performance and tracking systems, fatigue, game analysis, muscle fiber type composition, soccer

Section I: Physiology & Biochemistry

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1 | INTRODUCTION

Muscle fiber type composition has been regarded as a sports performance determinant ever since the classical studies in the 70s.[1,2] Moreover, it has a large genetic component[3,4] likely contributing to talent for sports with specific exercise demands. In explosive, short duration sports elite athletes typically have a large share of fast-twitch fibers (type IIa and IIx). These fibers have a higher maximal shortening velocity compared to slow-twitch fibers (type I), which contributes to larger power production capabilities.[5] On the other hand, the best endurance athletes dispose of a large proportion of fatigue resistant slow-twitch fibers (type I). Even within sports, muscle typology differences exist according to the discipline. For instance in world-class road cycling, athletes excelling in multistage races have a slower muscle typology than single-stage race specialists.[6,7] The optimal muscle fiber type composition in intermittent team sports is less straightforward, since these sports are characterized by mixed demands, including both prolonged low intensity movements as well as high-intensity actions.

In national level football (soccer), outfield players typically cover 10–13 km during a match. Most of this distance is covered by walking and low intensity running, but is interspersed with high-intensity runs, sprints, jumps, tackles, etc.[8] The reported percentages of fast-twitch fibers in these players' vastus lateralis muscle range from 35%–79%.[9,10] This indicates a large heterogeneity in muscle fiber type composition among professional football players. However, studies examining muscle fiber type composition in high level football players are scarce and sample sizes only go up to 15 players. Presumably because the current gold standard to measure muscle fiber type composition is a histological evaluation of muscle tissue obtained from a needle biopsy. This procedure has considerable limitations (e.g., sampling variance[11]) and is not feasible in large groups of high level athletes due to its invasive nature.

In 2011 a technique was validated to noninvasively estimate muscle fiber type composition by measuring muscle carnosine based on proton magnetic resonance spectroscopy (^{1}H-MRS).[12] Recently, Lievens et al. employed this method on a large cohort of high level football players and prospectively monitored hamstring strain injuries. Similar to the biopsy data, a wide variety of muscle typologies was found. Moreover, having a fast muscle typology was identified as a novel, and profound risk factor for hamstring strain injuries.[13] A potential reason why fast typology (FT) players are more prone to sustain a hamstring strain injury compared to slow typology (ST) players lies in the link between fatigue and hamstring strain injuries. Firstly, in football matches hamstring strain injuries are most frequently sustained at the end of each half when players experience fatigue.[14] Secondly, in lab conditions FT individuals have a more pronounced performance decrement during high-intensity exercise and need to recover longer compared to ST individuals.[15] Therefore, in the current study, it is of interest to investigate if FT players also fatigue more in a football setting. Mohr et al. examined this in either semiprofessional male or elite female football players during one experimental game,[16,17] but to our knowledge it has never been investigated in professional players during multiple official games, using real-life data.

Furthermore, little is known about the distribution of muscle typologies across players with different positional roles in the team. The physical game demands imposed on players are to some extent determined by their position. Therefore, players with a given muscle typology might perform better in a certain positional role. ST football players could be particularly suited for the central midfield position, since central midfielders cover the largest total distance,[18] but reach the lowest maximal speed.[19]

Besides evaluating on-field performance, clubs also test players' performance on off-field tests such as counter movement jump (CMJ), VO_{2peak} (peak oxygen consumption), and sprint tests to assess their physical strengths and weaknesses. Players with a fast sprinting time and high CMJ are identified as explosive players and the ones with a high VO_{2peak} as having an endurance profile. Currently it is unclear to what extent muscle typology underpins the performance on these tests.

Therefore, we aimed to further elucidate the importance of muscle typology in football by using the noninvasive ^{1}H-MRS-based technique in a large cohort of male professional football players. In this study we explored whether muscle typology is related to (1) the players' positional role, (2) running performance decrements during match-play by evaluating spatiotemporal tracking data, and (3) the players' performance on VO_{2peak}, CMJ and sprint tests.

2 | MATERIALS AND METHODS

2.1 | Participants

A total of 147 male international and national level football players (age, 24.7 ± 4.8 y) participated in this study.[20] Each player was part of the first team in one of four clubs competing in the Belgian Jupiler Pro League (first division). Subjects were categorized into the following positions based on a 4–3-3 formation: goalkeeper (N = 17); center back (N = 27); full back (N = 20); midfielder (N = 45); winger (N = 20); forward (N = 18). No distinction was made between attacking or defending midfielders.

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This study was approved by the local ethical committee (Ghent University Hospital; B670201837890) and all players gave their written consent to partake in this study.

2.2 | Study design

In the present observational study, all participants underwent a ^{1}H-MRS scan to have their muscle fiber typology estimated (Research question 1; Figure 1). Spatiotemporal data were collected during official games across four seasons to determine physical performance decrement defined as the decline in match running activities from the first to second half and were related to muscle fiber typology (Research question 2). Ten meter sprint performance, VO_{2peak} and CMJ performance were assessed and linked to the muscle fiber typology (Research question 3).

2.3 | Muscle fiber typology assessment

Carnosine content was measured in the soleus and gastrocnemius medialis muscle of each players' right leg by ^{1}H-MRS to estimate the muscle fiber type composition. All ^{1}H-MRS measurements were performed on a 3 T whole-body MRI scanner (Trio, Siemens) using the external referencing method as previously described by Lievens et al. (2021).[21]

The soleus and gastrocnemius carnosine concentrations were converted to z-scores relative to an age-matched control population of 163 healthy and recreationally active males. Afterwards, a mean z-score was calculated from the z-scores of the two muscles. Players were then assigned to one of three groups for their soleus, gastrocnemius and mean z-score: the ST group with a z-score ≤ −0.5, the intermediate typology (IT) group with a z-score between −0.5 and 0.5 or the FT group with a z-score ≥0.5.

2.4 | Match running output

Spatiotemporal data from four Jupiler Pro League teams were collected during 485 official games over four seasons (2017–2021) using electronic performance and tracking systems. Specifically, a semiautomatic video tracking SportVU system (Stats Perform, US) was used, which is a valid technology to analyze players' match running output.[22]

The distances that players covered during the full game, and during the first and second half were split into the following running intensity categories: strolling distance 0–0.99 km.h^{-1}; walking distance 1–5.99 km.h^{-1}; jogging distance 6–14.99 km.h^{-1}; running distance 15–19.99 km.h^{-1}; high speed distance 20–25 km.h^{-1}; high-intensity running distance >15 km.h^{-1}; sprinting distance >25 km.h^{-1}; medium acceleration distance 2–3 m.s^{-2}; high acceleration distance >3 m.s^{-2}; medium deceleration distance −2 to −3 m.s^{-2}; high deceleration distance <−3 m.s^{-2}.

Only data from outfield players who were on the pitch for at least 75 min during the game were retained and

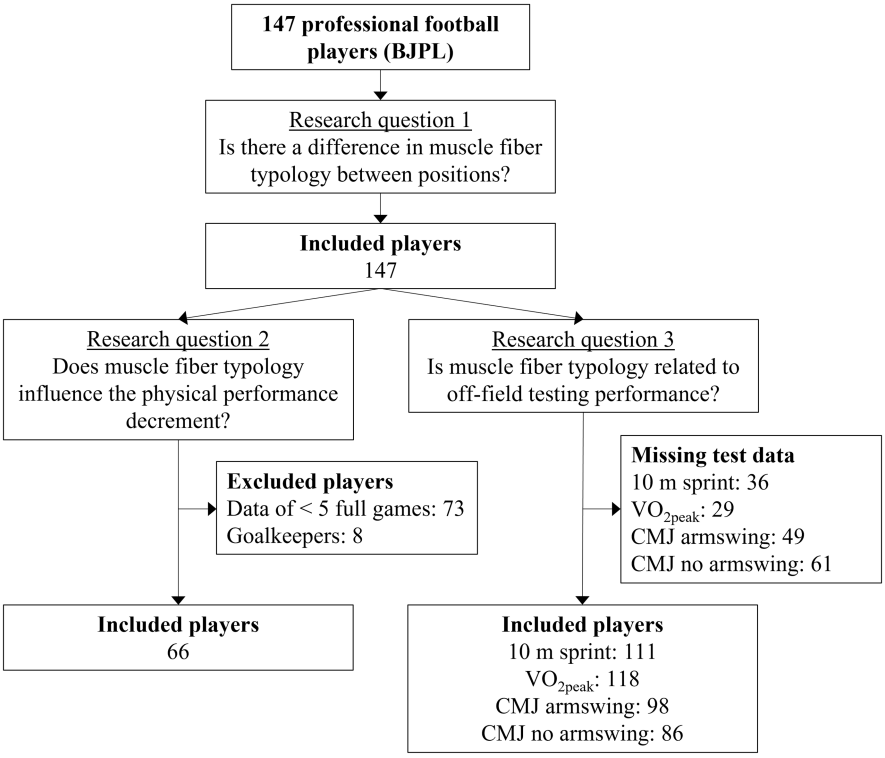

FIGURE 1 Study flowchart. For research question 2, full games are considered games in which a player played for at least 75 min. BJPL, Belgian Jupiler Pro League; VO2peak, peak oxygen consumption; CMJ, counter movement jump.

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covered distances were expressed per minute playtime. The players' average of at least five matches for covered distances during a full game, the first half and the second half in each intensity category were calculated to account for match-to-match variation. The decline in physical performance from the first to the second half was then determined for each running intensity category and served as a surrogate for fatigue toward the end of the game. On average we analyzed 35 ± 30 games (range: 5–137) per included player ($N = 66$).

2.5 | Off-field performance testing

Off-field physical performance was assessed in all clubs, however in one club the players did not perform CMJ.

2.5.1 | 10 m sprint

In two clubs subjects had to sprint for 10 m and the time was recorded using either double laser beam electronic timing equipment (Timing gates, Swift Performance) or infrared timing gates (Timing gates, Brower Timing System). The sprint was performed 2–3 times (depending on the club), and the best result was retained for data analysis.

In the other two clubs players performed 2–4 maximal sprints of 30 m, with split times at 5 m, 10 m, 20 m, and 30 m. The best 10 m split time was retained for data analysis (RaceTime2 chronometry and Polifemo light photocells; Microgate).

The 10 m sprint times in one club (2.07 ± 0.17 s) were markedly slower compared to the other three clubs (1.76 ± 0.08 s; 1.87 ± 0.06 s; 1.87 ± 0.12 s) due to slightly different measurement methods. Therefore, we calculated z-scores for each player by relating their performance to the mean and standard deviation of their own team's performance.

2.5.2 | Maximal incremental running test

The VO_{2peak} was assessed during preseason using a step-incremental maximal running test on a treadmill (Mercury). Two clubs used the protocol and metabolic cart which were previously described in Colosio et al. (2020).[23] Players from the other clubs started running at 5.4 $km.h^{-1}$ and speed was increased by 1.8 $km.h^{-1}$ every 3 min until volitional exhaustion, while pulmonary gas exchange was measured continuously (Cortex Metalyzer II and 3B). Given the similar VO_{2peak} results across clubs (56.3 ± 4.7 $mL.min^{-1}.kg^{-1}$; 57.8 ± 3.5 $mL.min^{-1}.kg^{-1}$; 58.5 ± 6.05 $mL.min^{-1}.kg^{-1}$; 58.6 ± 4.2 $mL.min^{-1}.kg^{-1}$), we did not calculate club specific z-scores.

2.5.3 | Counter movement jump

In three clubs subjects performed the CMJ with arm swing and in two clubs CMJ performance was also assessed without arm movement (i.e., with the hands on the hips). Subjects were instructed to jump as high as possible. The jump was performed 2–5 times (depending on the club) and the best result was preserved for data analysis. The jumping height was calculated from the flight time using either Optojump (Microgate) or FD4000s Dual force platform (ForceDecks). No club specific z-scores were calculated, since the CMJ performances without arm swing were comparable across clubs (41.0 ± 4.0 cm; 43.5 ± 3.9 cm). CMJ performances with arm swing in one club were worse than in two other clubs, but given the similar ranges (38.6–58.7 cm; 38.6–58.9 cm; 38.2–60.6 cm) we did not use z-scores.

2.6 | Statistical analysis

Data are presented as mean ± SD (standard deviation). Shapiro–Wilk test was performed to control for normality of the data and Levene's test to check if variances were equal across groups. One-way analyses of variance (ANOVA) were used to determine if soleus, gastrocnemius, and mean carnosine z-scores were different between playing positions. Additionally, we used chi-square tests to check if the relative amount of ST, IT, and FT players was different between positions.

One-way ANOVA were used to determine differences in distance covered in each running intensity category between typology groups for the full game, the first and second half. Differences in the decline in covered distances in each running intensity category from the first to second half between ST, IT, and FT players were also examined using one-way ANOVA. In case of significance, Tukey post hoc tests were applied to determine between which typology groups the covered distances was different. When the assumption of normality or equal variances were violated, a Kruskall–Wallis rank sum test with Wilcoxon post hoc test or a Welch ANOVA with Games-Howell post hoc test was employed, respectively.

Pearson's correlation tests were performed to assess the association between soleus, gastrocnemius and mean carnosine z-scores and off-field physical performance. Thresholds of 0.1, 0.3, 0.5, 0.7, and 0.9 were used for respectively small, moderate, large, very large, and extremely large correlation coefficients.[24] Differences in

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off-field physical performance between typology groups were assessed using one-way ANOVA. All statistical analyses were performed with R (version 4.2.0) and statistical significance was set at $p < 0.05$.

3 | RESULTS

3.1 | Muscle typology across different positional roles

On average the mean z-score was −0.09 ± 0.92 and our cohort consisted of 48 ST (33%), 67 IT (45%) and 32 FT players (22%). An overview of the mean z-scores and typologies for all six playing positions is given in Figure 2A,B. There were no differences in mean z-scores between positions ($p = 0.412$), nor in the relative amount of FT, IT, and ST players ($p = 0.891$). Similar results were found for the soleus and gastrocnemius z-scores ($p = 0.582$ and $p = 0.161$, respectively) and soleus and gastrocnemius typology groups ($p = 0.837$ and $p = 0.454$, respectively). It is clear that the range and variability in mean carnosine z-scores is large within each position (goalkeeper: −0.40 ± 0.87; midfielder: −0.17 ± 0.85; center back: −0.12 ± 0.90; forward: −0.07 ± 1.10; winger: 0.04 ± 0.93; full back: 0.22 ± 0.93) and that each typology is represented in all positions.

3.2 | Match running output

The relative distances covered in each intensity category per soleus typology group are given in Table 1 for the first half, the second half and the percentual decline from the first to second half. FT players covered 10.9% more high acceleration distance than ST players during the first half ($p = 0.021$) (Table 1; Figure 3A). No differences were found between FT, IT, and ST players for any other running intensity distances over the full game, and the first and second half (Table 1; Figure 3B,C).

However, the decline in covered high acceleration distance from the first to the second half was larger for FT (−0.36 ± 0.12 m.min^{-1}) players than ST (−0.21 ± 0.15 m.min^{-1}; $p = 0.006$) and IT players (−0.21 ± 0.19 m.min^{-1}; $p = 0.010$) (Figure 3D). Moreover, the decline in distance covered in several high-intensity zones tended to be larger in FT (high-intensity −2.98 ± 0.94 m.min^{-1}; high deceleration −0.37 ± 0.20 m.min^{-1}; medium acceleration − 0.76 ± 0.26 m.min^{-1}) than in ST players (high-intensity −1.88 ± 1.45 m.min^{-1}; high deceleration −0.22 ± 0.17 m.min^{-1}; medium acceleration − 0.51 ± 0.30 m.min^{-1}; $0.05 < p < 0.1$) (Figure 3E,F).

When categorizing muscle typology based on the gastrocnemius or on the mean (soleus + gastrocnemius), similar, yet slightly different results were found for match running output (see Tables SS1 and SS2). In short, for the gastrocnemius typology groups, the decline in covered high acceleration distance was larger in FT than in IT players (−0.32 ± 0.18 m.min^{-1} vs. -0.19 ± 0.18 m.min^{-1}, respectively; $p = 0.018$). For the mean z-score, ST and IT differed for the decline in covered distance from the first to second half in the jogging intensity zone (−6.27 ± 1.79 m.min^{-1} vs. -4.86 ± 1.82 m.min^{-1}, respectively; $p = 0.038$) and in total distance (−7.16 ± 1.35 m.min^{-1} vs. −5.82 ± 1.82 m.min^{-1}, respectively; $p = 0.046$).

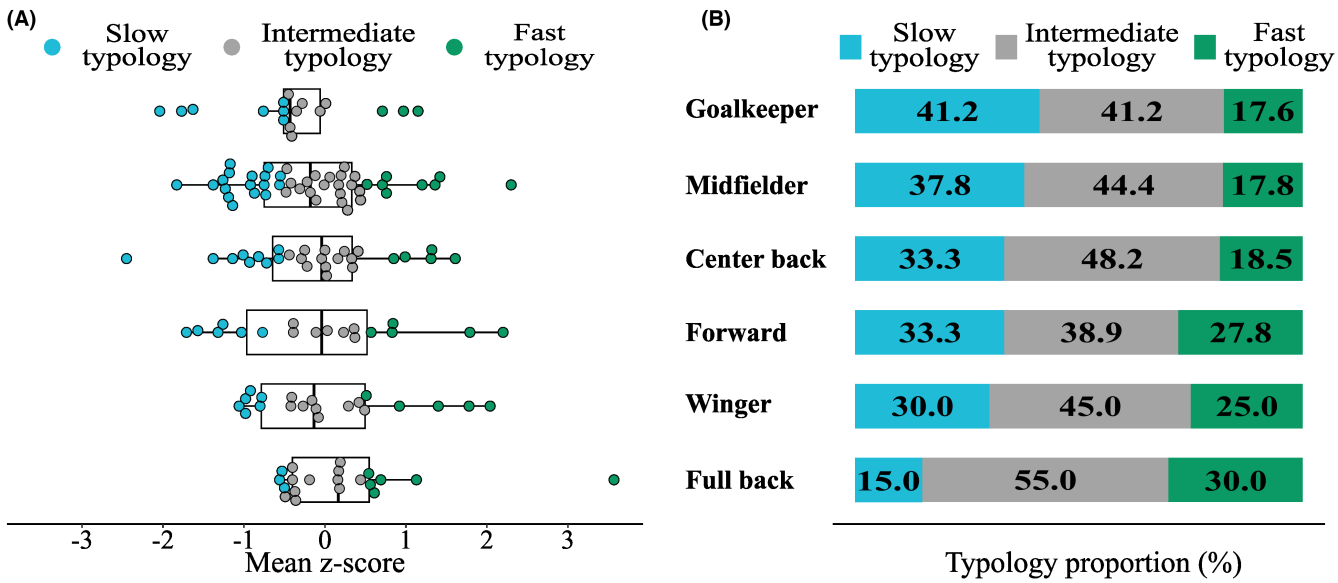

FIGURE 2 Muscle typology in national level football players per position. (A) Boxplots and individual mean z-scores per position. The color of each circle represents the mean typology. (B) Relative amount of ST, IT and FT players in each position.

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TABLE 1 Overview of the relative distances covered by ST, IT and FT players based on soleus z-score in the studied intensity zones during the first half, second half and the change from the first to second half.

	Relative covered distance during first half ($m.min^{-1}$)			Relative covered distance during second half ($m.min^{-1}$)			Change in relative covered distance from first to second half (%)		
Running intensity zone	**ST**	**IT**	**FT**	**ST**	**IT**	**FT**	**ST**	**IT**	**FT**
Total	118.0 ± 7.76	118.1 ± 7.81	120.1 ± 6.12	111.8 ± 6.81	111.9 ± 7.72	112.9 ± 7.00	−5.19 ± 1.55	−5.24 ± 1.52	−6.07 ± 1.62
Strolling (0–0.99 $km.h^{-1}$)	0.46 ± 0.07	0.46 ± 0.08	0.44 ± 0.08	0.56 ± 0.09	0.55 ± 0.09	0.54 ± 0.08	21.01 ± 5.34	20.8 ± 10.24	24.83 ± 6.41
Walking (1–5.99 $km.h^{-1}$)	29.72 ± 1.74	30.15 ± 1.93	30.43 ± 2.22	30.77 ± 1.42	31.15 ± 1.59	31.41 ± 2.50	3.64 ± 3.35	3.46 ± 3.47	3.2 ± 2.11
Jogging (6–14.99 $km.h^{-1}$)	62.52 ± 5.33	61.4 ± 5.55	61.96 ± 5.66	57.08 ± 4.11	56.26 ± 4.81	56.61 ± 6.14	−8.57 ± 2.95	−8.3 ± 2.64	−8.72 ± 3.03
Running (15–19.99 $km.h^{-1}$)	16.38 ± 3.81	16.57 ± 3.39	17.4 ± 2.82	15.08 ± 3.27	15.20 ± 3.26	15.47 ± 3.07	−7.55 ± 4.97[§]	−8.26 ± 5.23	−11.48 ± 4.92
High speed (20–25 $km.h^{-1}$)	6.66 ± 1.69	7.08 ± 1.70	7.37 ± 1.18	6.24 ± 1.43	6.52 ± 1.68	6.59 ± 1.18	−5.26 ± 9.92	−7.93 ± 6.84	−10.70 ± 4.95
Sprint (> 25 $km.h^{-1}$)	2.20 ± 0.84	2.45 ± 1.03	2.51 ± 0.80	2.04 ± 0.77	2.23 ± 0.99	2.24 ± 0.70	−6.03 ± 12.79	−8.74 ± 10.97	−10.15 ± 8.60
High intensity (> 15 $km.h^{-1}$)	25.24 ± 5.61	26.1 ± 5.54	27.28 ± 3.68	23.37 ± 4.85	23.96 ± 5.41	24.3 ± 4.00	−7.05 ± 5.30[§]	−8.31 ± 4.84	−11.21 ± 4.19
Medium acc (2–3 $m.s^{-2}$)	6.23 ± 0.85	6.51 ± 1.00	6.72 ± 0.40	5.72 ± 0.78	5.97 ± 0.97	5.96 ± 0.58	−8.11 ± 4.76	−8.33 ± 5.16	−11.45 ± 4.41
High acc (> 3 $m.s^{-2}$)	2.62 ± 0.41*	2.80 ± 0.55	2.91 ± 0.18	2.41 ± 0.37	2.59 ± 0.50	2.54 ± 0.19	−7.69 ± 5.93[§]	−7.30 ± 6.27*	−12.42 ± 3.94
Medium dec (−2 – −3 $m.s^{-2}$)	5.89 ± 0.87	6.17 ± 1.04	6.24 ± 0.53	5.39 ± 0.73	5.64 ± 0.96	5.57 ± 0.59	−8.26 ± 4.75	−8.43 ± 4.71	−10.87 ± 4.61
High dec (< −3 $m.s^{-2}$)	2.63 ± 0.47	2.82 ± 0.60	2.84 ± 0.36	2.41 ± 0.40	2.56 ± 0.51	2.48 ± 0.32	−8.08 ± 6.15[§]	−8.61 ± 6.06	−12.73 ± 6.38

Note: *Values are statistically different ($p < 0.05$) from the FT within the same intensity zone and period. [§]Superscripts indicate trends for statistically different values ($0.05 < p < 0.01$) from the FT within the same intensity zone and period.

Abbreviations: acc, accelerations; dec, decelerations; IT, intermediate typology players; FT, fast typology players; ST, slow typology players.

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3.3 | Off-field performance tests

The relative VO_{2peak}, 10 m sprint and CMJ performances are depicted per mean typology group in Figure 4A–D. FT players jumped higher than ST players when performing a CMJ with arm swing (52.1 ± 4.2 vs. 48.2 ± 5.6 cm, respectively; $p = 0.024$) and without arm swing (45.2 ± 2.8 vs. 40.6 ± 4.2 cm, respectively; $p < 0.001$), while CMJ performance of IT players was not statistically different from ST or FT players. Moreover, the IT players tended to perform better than ST players on the CMJ with arm swing (49.2 ± 5.0 for IT; $p = 0.097$), but worse than FT ($p = 0.063$). No differences were found between groups for the VO_{2peak} ($p = 0.313$) and the 10 m sprint ($p = 0.142$).

Correlations between the muscle typology z-scores and off-field performance are given in Table 2. There were small to moderate positive correlations between mean, soleus and gastrocnemius muscle typology and the CMJ without arm swing performance ($r = 0.23$–0.36; $p < 0.05$). Furthermore, gastrocnemius muscle typology, but not mean and soleus typology, was negatively associated with the 10 m sprint z-score ($r = -0.22$; $p = 0.026$). Muscle typology was not correlated with VO_{2peak} and CMJ performance with arm swing.

4 | DISCUSSION

In the present study we examined the muscle typology in 147 professional football players using ^{1}H-MRS. Our findings revealed that, on average, players had an intermediate muscle typology, however with large inter-individual differences. There were no significant differences in muscle typology between players occupying different positions. Although, muscle typology did affect the players' high-intensity running activities near the end of the game. Fast soleus typology players had a larger decline in distances at high intensities from the first to second half compared to IT and ST players. Furthermore, FT players outperformed ST players on the CMJ, but 10 m sprint and VO_{2peak} performances were not different across groups.

The current study demonstrated a large muscle typology heterogeneity among professional football players. Mean carnosine z-scores ranged from −2.45 to 3.57, encompassing muscle typologies usually found in elite ironman triathletes and marathon runners at the slow-twitch end and those of elite track-and-field and track cycling sprinters at the fast-twitch end of the spectrum.[6,7] This is in agreement with the large range in muscle biopsy-based fiber type composition observed among national level football players in previous low sample size studies.[9,10]

In a preliminary report on 118 high level football players, no differences in muscle typology were found between positions (goalkeepers, defenders, midfielders and attackers).[13] The current more in-depth analysis using six positional roles also showed no muscle typology differences between positions. Moreover, there were ST, IT, and FT players in all positions and the within-position muscle typology variation was large. These findings suggest that having a certain muscle typology is not required to meet position-specific demands in the Belgian first division. However, it was remarkable that 41.2% of the goalkeepers were ST players. Goalkeepers typically only cover 5–6 km per game while walking 73% of the time and some of their explosive actions (i.e., saves and clearances) are the most decisive for the match result.[25] Hence, we expected goalkeepers to have a faster muscle typology compared to outfield players.

Multiple studies have shown a decline in acceleration and deceleration distance,[26] and high-intensity running[27,28] toward the end of professional football games. We aimed to clarify if muscle characteristics could explain the degree of this decline. Indeed, FT players in our study exhibited a larger decrement from the first to the second half in relative high acceleration distance (>3 m.s^{-2}) compared to ST and IT players. Their respective percentual decreases corresponded to 12.4% (FT), 7.7% (ST), and 7.3% (IT). Moreover, FT players tended to have larger declines in distances covered high-intensity running (> 15 km. h^{-1}), accelerating at 2–3 m.s^{-2} and decelerating at < −3 m.s^{-2} compared to ST players ($0.05 < p < 0.10$). Overall, it seems that during the second half FT players cannot sustain their first half physical performance in these high-intensity zones as well as ST players. This finding is consistent with research conducted by Mohr and colleagues in elite female and semiprofessional male players. They have shown that skeletal muscle phenotype markers for endurance, mainly myosin heavy chain type I protein expression and beta-hydroxyacyl-CoA-dehydrogenase activity, are associated with high-intensity running distance in the last 15 min of a game.[16,17] A larger amount of high-intensity running during the first half negatively impacts running performance in the second half.[29] Hence, it could be argued that FT players have a larger reduction in high-intensity running activities due to their higher physical performance during the first half, but in the present study this was only the case for high acceleration distance. Therefore, we expect that fatigue toward the end of the game is more pronounced in FT players. Still, the fact that FT players covered more high acceleration distance during the first half could indicate their potential for powerful actions early in the game. These actions are crucial during the most decisive plays in a match.[30]

Whether the decline in high-intensity running activities originates from fatigue, pacing strategies, specific game circumstances or tactics could be debated.[29]

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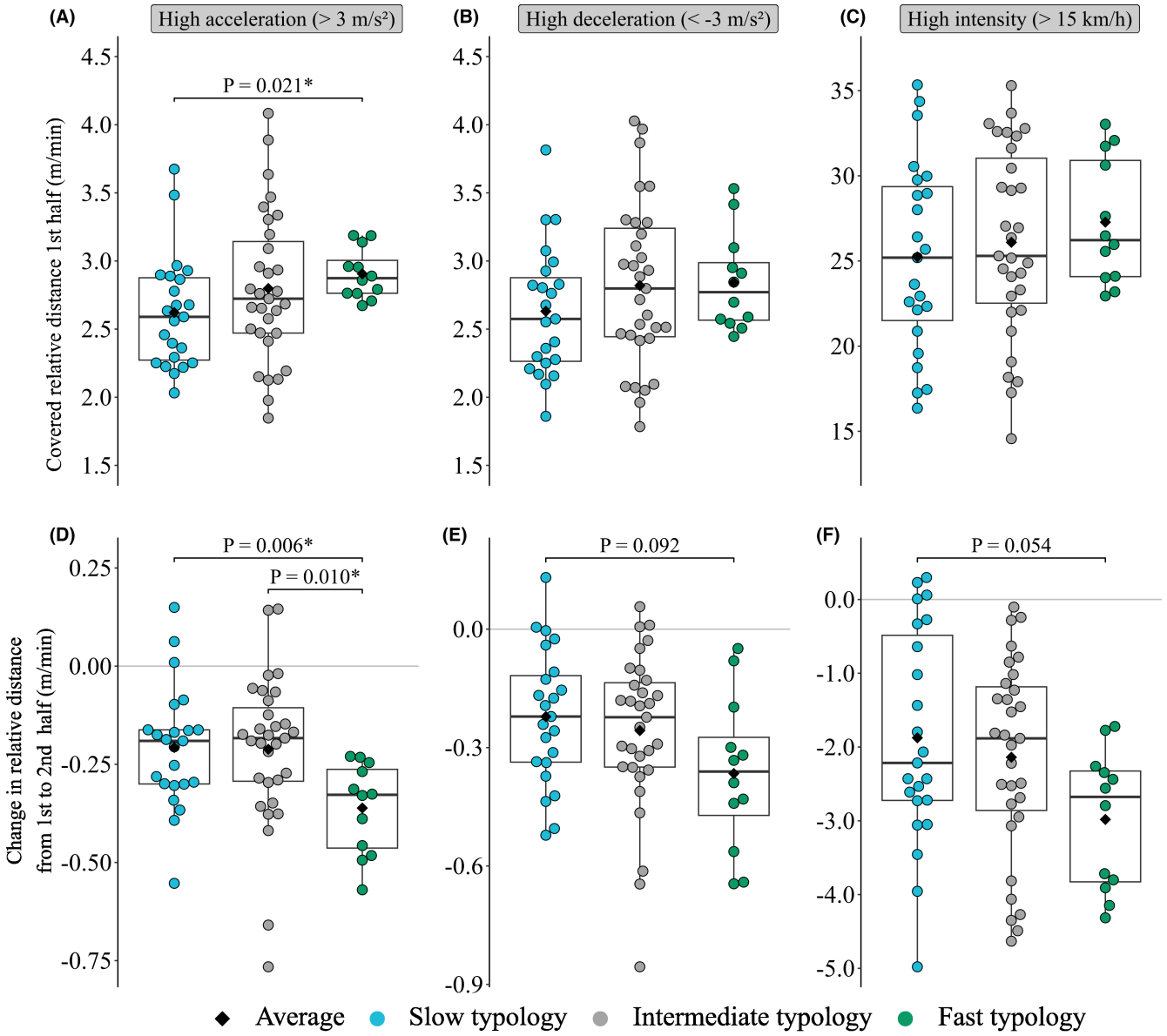

FIGURE 3 Relative covered distances during the first half in the high acceleration (A), high deceleration (B) and high intensity (C) zones per soleus typology group. The decline in relative covered distances from the first to the second half in the high acceleration (D), high deceleration (E) and high intensity zone (F) for ST, IT, and FT players. The color of each circle represents the soleus typology and black diamonds indicate the average relative covered distance per group. *Values indicate statistical differences between soleus typology groups ($p < 0.05$).

Previous research has shown that game interruptions increase toward the end of a match and that this results in a decrease in match running output.[31] Accordingly fatigue at the whole group level could be somewhat overestimated because also tactical time killing contributes to the decline in running output. However, studies have shown that physical performance assessed by numerous tests such as repeated sprints[32] and CMJ[33] is reduced post-match, which confirms the manifestation of fatigue during the later stages of a football match. The fact that substitutes cover more high-intensity running during the last 15 min of a game compared to starters who play 90 min or more further substantiates the fatigue-induced decrement in high-intensity movements toward the end of the game.[34]

Aside from investigating performance on the pitch, high level football clubs also regularly perform physical performance tests off the pitch to identify players' strengths and weaknesses. Although our study was not specifically designed to answer this research question, we evaluated if muscle typology was an underpinning factor for the players' off-field performance. This seems to be the case for the CMJ, since muscle typology was correlated with the players' CMJ performance without arm swing ($r = 0.23–0.36$; $p < 0.05$) and FT players

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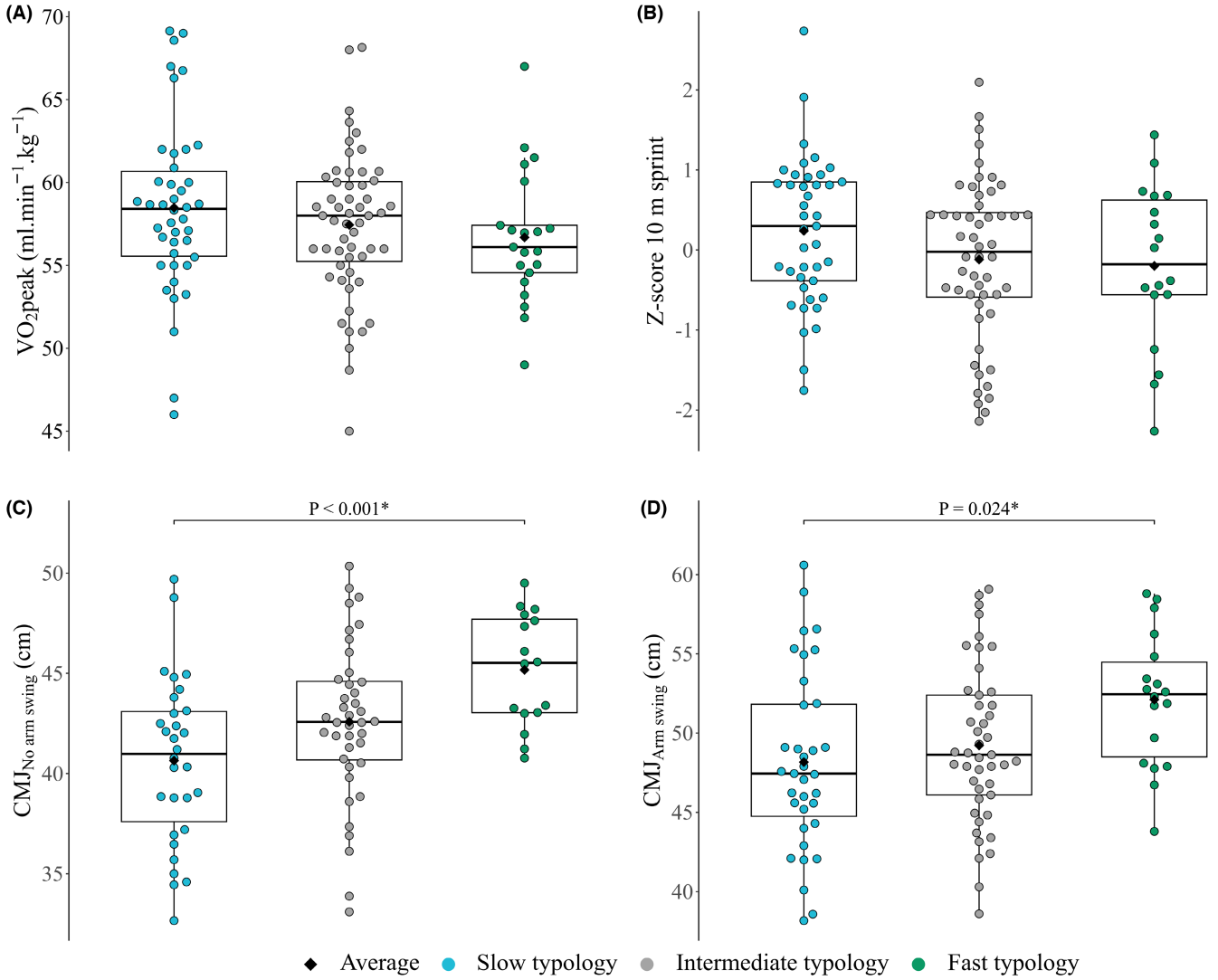

FIGURE 4 Off-field performance per mean typology group. The relative VO_2peak (A), 10 m sprint performance (lower z-scores indicate faster sprint times) (B) and counter movement jump height without (C) and with (D) arm swing are illustrated. The color of each circle represents the mean typology and black diamonds indicate the average performance per group. Abbreviations; VO_{2peak}, peak oxygen consumption; CMJ, counter movement jump. *Values indicate statistical differences between groups ($p < 0.05$).

TABLE 2 Pearson correlations between muscle typology and off-field performance test parameters.

Muscle typology (z-score)	VO_{2peak} ($ml.min^{-1}.kg^{-1}$)	10 m sprint time (z-score)	$CMJ_{No\ arm\ swing}$ (cm)	$CMJ_{Arm\ swing}$ (cm)
Mean	−0.15	−0.14	0.31*	0.17
Soleus	−0.09	−0.10	0.36*	0.19
Gastrocnemius	−0.17	−0.22*	0.23*	0.14

Note: *Values are significant correlations ($p < 0.05$).

Abbreviations: $CM_{No\ arm\ swing}$, counter movement jump without arm swing; $CMJ_{Arm\ swing}$, counter movement jump with arm swing; VO_{2peak}, peak oxygen consumption.

outperformed ST players. A possible explanation why CMJ performance with arm swing was not significantly correlated with muscle typology might be that it involves more coordination and relies less on pure lower limb explosiveness compared to the CMJ without arm swing. For VO_{2peak}, we did not find differences between typology groups, but to reach high VO_{2peak} levels it is likely still beneficial to have a slow muscle typology. Since 14% of ST players had a VO_{2peak} greater than 65 $mL.min^{-1}.kg^{-1}$, whereas this was only the case for 4% and 5% of

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IT and FT players, respectively. Ten meter sprint performance was also not different between groups, meaning that players who are identified as explosive based on 10 m sprint time, do not necessarily possess a fast-twitch muscle profile based on the ^{1}H-MRS technique and vice versa.

There was a good agreement between the carnosine concentrations in the players' soleus and gastrocnemius muscles ($r = 0.74$; $p < 0.001$), but it was not perfect. Therefore, some players who were ST or FT based one muscle could be IT in the other one. As a result the answers on the research questions vary slightly depending on assessing either the soleus, gastrocnemius or mean muscle typology. For research question 1 and 3 there were only minor differences between the three ways of categorization, but for research question 2 the largest effects were clearly observed when grouping for soleus typology. Presumably this relates to a superior reproducibility of the ^{1}H-MRS carnosine measurement in the soleus compared to the gastrocnemius muscle (test–retest coefficient of variation 4.3% vs. 7.6%, respectively, see Ozdemir et al., 2007).[35]

One limitation of this study is that we did not account for differences in playing position or physical fitness in the match running output analysis. Given the different physical demands in different tactical roles, this may have affected the results. However, as previously shown, even within positions large muscle typology differences exist between players. Moreover, the decrement in high-intensity running activities was scaled to the individual as it is a comparison of the player's physical performance in the first and second half. The fact that we did not consider contextual game factors is a second limitation, because it has been shown that winning or losing, playing home or away games and the strength of the opposing team influences players' match running. For instance, Lago and colleagues found that professional players in La Liga performed less running at speeds faster than 19.1 km. h^{-1} when winning as compared to losing.[36] To minimize the influence of these contextual factors in addition to red cards, large goal differences and importance of the game, we averaged at least five match observations per player. One could argue that five games per player is not sufficient to level out the effect of contextual variables, so we reanalyzed the match running data with a more strict inclusion criterium of at least 10 match observations. Hereby, the number of included players was reduced from 66 to 56. The results were nearly identical to the initial analysis, leading the authors to conclude that either five games was enough to account for the contextual factors or that their influence was limited in the current dataset. A third limitation was that the off-field testing procedures were not standardized across clubs. This could have impacted the comparison between players of different clubs. However, the results already give a first impression about the relationship between muscle typology and off-field performance in football players, which can be investigated more thoroughly in future research.

5 | CONCLUSION

The current study revealed a large variety in muscle typologies among national level football players, even within playing positions. Additionally, we demonstrated that the decrement in high-intensity running activities from the first to second half is more pronounced in FT than ST players. This suggests that FT might fatigue more toward the end of a game compared to ST.

6 | PERSPECTIVES

Since the current results indicate that the positional role of professional football players is independent of their muscle typology, we would not advise coaches to orient players toward a certain position based on muscle typology.

FT players had a more pronounced decrement in high-intensity movements from the first to second half compared to ST players, which suggests a more pronounced fatigue in FT players toward the end of the game. Therefore, high level coaches should consider their players' muscle typology when making tactical decisions. For instance, having worthy substitutes for the positions in which FT players are starting could be of particular importance.

Coaches should also be aware of FT players whose high-intensity running activities are deteriorating during games, training camps, and heavy training sessions as this might indicate an elevated muscle injury risk.[13] Realtime monitoring of spatiotemporal data could help in detecting physical performance decrements.

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CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflict of interest.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONSENT

All subjects gave their written informed consent to take part in the study.

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